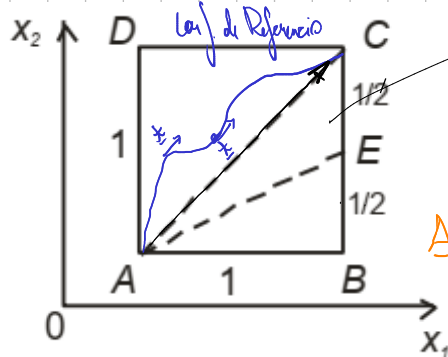
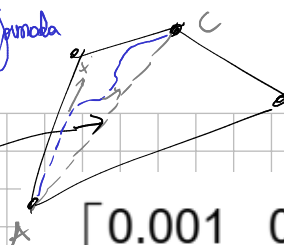




Conf. Deformada

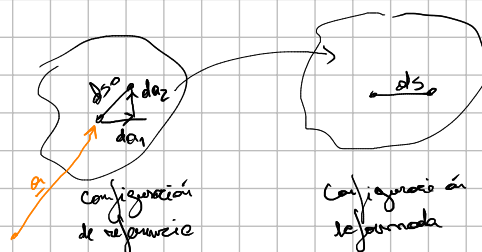


$$\oint f(s) ds = \int_1 \int_1 f(x_1, x_2) \sqrt{x_1^2 + x_2^2} dx_1 dx_2$$

$$\begin{bmatrix} 0.001 & 0.002 & 0.003 \\ 0.002 & 0.001 & 0 \\ 0.003 & 0 & 0.002 \end{bmatrix}$$

$$AE = \begin{bmatrix} 1 \\ 1/2 \\ 0 \end{bmatrix}$$

$$ds^2 - ds_0^2 = 2E_{ij} da_i da_j$$



diferencial

$$ds^2 - ds_0^2 = (ds + ds_0)(ds - ds_0) \Rightarrow ds - ds_0 = E_{ij} \frac{da_i}{ds_0} \frac{da_j}{ds_0} ds_0$$

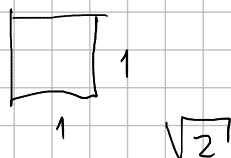
longitud de curva

$$\int_S ds - ds_0 = \int_0^{l_0} \frac{ds - ds_0}{d\Delta l} d\Delta l = \Delta l = \int_0^{l_0} E_{ij} t_i t_j ds_0 = E_{ij} t_i t_j \int_0^{l_0} ds_0$$

campo de curvatura

$$\Delta l = \underline{t} \cdot \underline{E} \cdot \underline{t} \cdot l_0$$

$$= \frac{l_0 \cdot \underline{t} \cdot \underline{E} \cdot l_0 \cdot \underline{t}}{l_0} =$$



AC

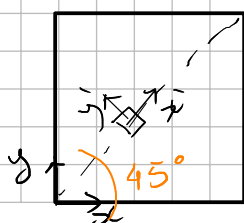
$$\underline{AC} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \Delta l = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0.001 & 0.002 & 0.003 \\ 0.002 & 0.001 & 0 \\ 0.003 & 0 & 0.002 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} / \sqrt{2}$$

$$\Delta l = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0.003 \\ 0.003 \\ 0.003 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} = \frac{0.006}{\sqrt{2}} = \frac{2 \cdot 0.003}{\sqrt{2}} = 0.003 \sqrt{2}$$

$$E \rightarrow E'$$

$$\rightarrow F_{x'x'} \cdot l = \Delta l$$

$$F_{x'x'} = \frac{\partial \sigma_{x'}}{\partial x'}$$



Versión 2:

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta,$$

$$\begin{bmatrix} 0.001 & 0.002 & 0.003 \\ 0.002 & 0.001 & 0 \\ 0.003 & 0 & 0.002 \end{bmatrix}$$

$$\begin{aligned} \epsilon_{x'x'} &= \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2} \cos(2 \cdot 45^\circ) + \tau_{xy} \sin(2 \cdot 45^\circ) \\ &= \frac{0,001 + 0,001}{2} + 0,002 = 0,003 \end{aligned}$$

$$\Delta l = \epsilon_{x'x'} \cdot l_0 = 0,003 \cdot \sqrt{2}$$

Pequeñas deformaciones y campo uniforme de deformaciones

Versión 3:

$$\epsilon_{xx} = \frac{\partial u_x(x,y)}{\partial x} \Rightarrow du_x(x,y) = \epsilon_{xx} dx \Rightarrow u_x(x,y) = \epsilon_{xx} \int dx + C_1(y) = \epsilon_{xx} \cdot x + C_1(y)$$

$$\epsilon_{yy} = \frac{\partial u_y(x,y)}{\partial y} \Rightarrow du_y(x,y) = \epsilon_{yy} dy \Rightarrow u_y(x,y) = \epsilon_{yy} \int dy + C_2(x) = \epsilon_{yy} \cdot y + C_2(x)$$

$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) = \frac{1}{2} \left(\frac{\partial C_1(y)}{\partial y} + \frac{\partial C_2(x)}{\partial x} \right)$$

$$\int \epsilon_{xy} dx = \frac{1}{2} \left(\frac{\partial C_1(y)}{\partial y} \cdot \int dx + \int \frac{\partial C_2(x)}{\partial x} dx \right) + C_3$$

$$\epsilon_{xy} x = \frac{1}{2} \left(\frac{\partial C_1(y)}{\partial y} x + C_2(x) \right) + C_3 \Rightarrow$$

$$C_2(x) = 2\epsilon_{xy} \cdot x - \frac{\partial C_1(y)}{\partial y} \cdot x - 2C_3$$

Si se integra con respecto a y $\Rightarrow \epsilon_{xy} \cdot y = \frac{1}{2} \left(C_1(y) + \frac{\partial C_2(x)}{\partial x} \cdot y \right) + C_4$

$$\epsilon_{xy} \cdot y = \frac{1}{2} \left(C_1(y) + 2\epsilon_{xy} \cdot y - \frac{\partial C_1(y)}{\partial y} \cdot y \right) + C_4 \Rightarrow C_1(y) - \frac{\partial C_1(y)}{\partial y} \cdot y + 2C_4 = 0$$

Solución de la ecuación diferencial en la función $C_1(y)$

$$\Rightarrow C_1(y) = C_5 y - C_4 \quad \Rightarrow C_2(x) = 2\epsilon_{xy} \cdot x - C_5 x - C_3$$

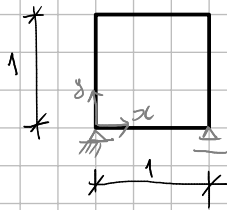
$$u_x = \epsilon_{xx} \cdot x + C_5 \cdot y - C_4$$

3 incógnitas: C_3, C_4 y C_5

$$u_y = \epsilon_{yy} \cdot y + 2\epsilon_{xy} \cdot x - C_5 x - C_3$$

$$\text{si } C_5 = \epsilon_{xy} + C_5 \Rightarrow \begin{cases} u_x = \epsilon_{xx} \cdot x + (\epsilon_{xy} + C_5) \cdot y - C_4 \\ u_y = \epsilon_{yy} \cdot y + (\epsilon_{xy} - C_5) \cdot x - C_3 \end{cases}$$

Para restringir el movimiento rígido se utiliza:



$$u_x(0,0) = 0$$

$$u_y(0,0) = 0$$

$$u_y(1,0) = 0$$

$$u_x(0,0) = \epsilon_{xx} \cdot 0 + \epsilon_{xy} \cdot 0 - \epsilon_4 = 0 \Rightarrow \epsilon_4 = 0$$

$$u_y(0,0) = \epsilon_{yy} \cdot 0 + 2 \cdot \epsilon_{xy} \cdot 0 - \epsilon_5 \cdot 0 - \epsilon_3 = 0 \Rightarrow \epsilon_3 = 0$$

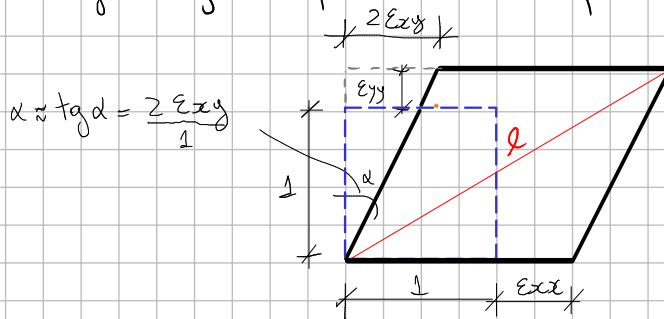
$$u_y(1,0) = \epsilon_{yy} \cdot 0 + 2 \epsilon_{xy} \cdot 1 - \epsilon_5 \cdot 1 - 0 = 0 \Rightarrow \epsilon_5 = 2 \epsilon_{xy}$$

ángulo de deformación

$$u_x = \epsilon_{xx} \cdot x + 2 \epsilon_{xy} \cdot y$$

$$u_y = \epsilon_{yy} \cdot y + 2 \cancel{\epsilon_{xy} \cdot x} - 2 \cancel{\epsilon_{xy} \cdot x} = \epsilon_{yy} \cdot y$$

Notar que al ser uniforme la deformación, la deformación de la estructura se puede aproximar al comportamiento de un cubo diferencial y por la interpretación de los componentes de tensor de deformación:



$$\alpha \approx \tan \alpha = \frac{2 \epsilon_{xy}}{1}$$

$$l = \sqrt{(2 \epsilon_{xy} + 1 + \epsilon_{xx})^2 + (1 + \epsilon_{yy})^2}$$

$$= \sqrt{(2 \cdot 0,002 + 1 + 0,001)^2 + (1 + 0,001)^2} = \sqrt{(1 + 0,005)^2 + (1 + 0,001)^2}$$

$$= \sqrt{1 + 0,010 + 0,005^2 + 1 + 0,002 + 0,001^2} = \sqrt{2 + 0,012 + 0,000026}$$

$$= \sqrt{2} \left(\sqrt{1 + 0,006 + 0,000013} \right)$$

$$\Delta l = l - l_0 = \sqrt{2} \left(\sqrt{1 + 0,006 + 0,000013} - 1 \right)$$

0,000003

para que de exactamente 0,003

La diferencia está en la dirección dónde se integra y por lo tanto la dirección que se aplica la aproximación de pequeños deformaciones (en x e y o sobre la diagonal).